

Project Report

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My project begins with getting acquainted with tools. With the help of ASIO, the audio was manipulated in code for the first time.

In Project0, we completed recording and playback, and became familiar with multi-channel functions and operations. The Tutorial on Github provides a basic framework for performing operations by modifying the 'audioDeviceIOcallbacks' function. After setting the frequency, amplitude, sampling rate, and constructing a sine function, use "OutputChannelData" to output multi-channel audio. At the same time, set a timer in the main program to switch output (specific audio or recording).

In Project 1, we were introduced to the sound card kit. Starting from specific frequency sound waves, we attempt to use sound waves to carry information. Transmitting 01 signal between two nodes through PSK modulation. PSK modulation transmits information by changing the phase of the carrier wave. Specifically, PSK modulation controls the phase of the carrier through digital baseband signals, and the phase changes of the carrier correspond to different data. For example, in BPSK (Binary Phase Shift Keying), binary data "0" and "1" correspond to the 0 degree and 180 degree phase changes of the carrier, respectively. The signal propagates through an analog PHY frame, consisting of a preamble, length field, and cyclic redundancy check. The preamble is used for frame synchronization, helping the receiver identify the starting position of the frame and perform clock synchronization. It usually consists of a series of specific bit sequences (such as 101010...) that are sent at the beginning of data transmission. The length field represents the length of the

subsequent data portion (i.e. payload). The length of this field can vary depending on the protocol, usually represented by a fixed length field to indicate the number of bytes in the data portion. CRC is a verification mechanism used to detect whether errors occur in data during transmission. By applying the CRC algorithm to the data section, a fixed length checksum is generated and attached to the end of the frame. The sender calculates the CRC value of the data portion and attaches it to the frame; After receiving the data, the receiver also calculates the CRC value and compares it with the received CRC. If both are consistent, the data is valid; Otherwise, it indicates that there was an error in the data transmission process and a request for retransmission is required.

In Project 2, we use data cables to transmit signals. Compared to Project 1, this approach has less noise and higher operability for sound waves. In addition, ACKs are also used to improve the reliability of transmission. The sender encapsulates the data into frames (containing data content and control information), and then sends the data to the receiver through the network. After receiving the data frame, the receiver first checks whether the data is complete and correct (CRC check or other forms of error checking may be performed). If the data is complete and error free, the receiver will send an ACK message. The recipient's ACK message contains an acknowledgement of the received data. For example, it may contain the sequence number of the data frame, indicating which frame the received data belongs to. ACK typically includes an acknowledgement sequence number, indicating which data frame the receiver successfully received and processed. The receiver can inform the sender of the sequence number of the next frame they expect to receive through an acknowledgement number (such as the acknowledgement number in TCP protocol). This means that the sender can start sending frames with higher sequence numbers. After receiving the ACK, the sender confirms that the data has been successfully delivered to the receiver and usually begins sending the next part of the data. If the sender does not receive an ACK, or if the ACK received does not match the expected one, it will usually resend the data (retransmission mechanism). If no ACK is received within the specified time (such as timeout), the sender will resend the data frame.

This mechanism ensures that communication can continue even in the event of data loss or transmission errors.

In Project 3, we attempted to use the campus network to 'ping' another device on one device. ICMP Echo consists of two message types: Echo Request and Echo Reply. These two types of messages are commonly used in the ICMP protocol to detect whether the target host is online and whether the connection between them is available. Echo Request is sent from the source host to the target host, with the purpose of requesting the target host to return a response to confirm whether the network connection is normal. The main information included is the identifier (ID), serial number, and data payload. Identifiers and serial numbers are typically used to match the pairing of requests and responses. Echo Reply is a response from the target host to the Echo Request message, indicating that the target host is reachable and responding normally. The echo response message will carry the identifier, serial number, and data payload sent by the source host, helping the sender match each request with the response. Then simulate a router "ping" another device under a hotspot and become familiar with NAT. The basic working principle of NAT is that when the computer in the private network initiates communication to the Internet, the NAT device (usually the router) will convert the private IP address to the public IP address, and then convert the public IP address back to the corresponding private IP address when the communication returns.

In Project 4, we became familiar with DNS and HTTP operations, similar to Project 3 which uses "ping" to verify connections. Connect one device to the campus network and separate another device from the campus network, using DNS to make it possible for the latter to connect.

It has to be said that projects are quite challenging. Especially during the debugging phase. I once exported all the content in 'OutputChannelData' and compared it line by line to try to find errors. This is undoubtedly very time-consuming, but it is difficult to find a more direct debugging method beyond that. Also, it is recommended to update the Tutorial and Matlab files on Github. The

Tutorial version is outdated compared to the version used this semester, and the Matlab files may even be unable to run. In addition, can data transmission between multiple devices be achieved through other means? This semester, I am working as a solo team, and local debugging is done between a laptop and a desktop computer. And Project's check needs to be conducted in the classroom, which has caused me trouble. I even invited a teaching assistant to my dormitory for a check at one point. Desktop computers are really inconvenient to carry, and the code running efficiency and transmission noise vary between different devices. I have tried borrowing other classmates' devices, but it will lead to a decrease in the efficiency of the final result. Due to the impact of the postgraduate entrance examination, I regret that I was unable to complete all the tasks of the project. However, I am still very grateful for the help of the teachers and teaching assistants, which allowed me to achieve the requirements of the project as much as possible. I also gained a lot of fun from it. As we build our own computer network step by step, we also constantly gain a sense of achievement.